

2008 JC2 [H2] Prelim II Paper 3 (Answers)

Question 1

(a)(i)

via gel electrophoresis;

DNA restriction fragments obtained using restriction enzyme loaded into special wells on agarose gel placed in chamber containing buffer solution;

electric field or voltage applied across gel;

negatively charged restriction fragments migrate, from wells, through gel matrix towards positive electrode / anode;

DNA fragments thus separated by their size / length;

gel matrix acts as a molecular sieve, allowing smaller / shorter restriction fragments to migrate to opposite end faster;

(a)(ii)

via nucleic acid hybridization (Southern blotting), using a specific radioactive, single-stranded probe, complementary to targeted DNA sequence;

d/s DNA fragments on the gel is also denatured in alkaline solution;

blotting of s/s DNA onto nitrocellulose or nylon membrane;

s/s labeled probe will hybridise via hydrogen-bonds with targeted, complementary DNA sequence;

detected by autoradiography as dark bands on photographic / X-ray films;

(b)(i)

mother (A) is a carrier, carries a normal allele (allele 2) on one X-chromosome and a defective, recessive allele on the other which is not visible on profile as probe not able to bind due to deletion in defective gene;

father (B) carries only a copy of the normal allele (allele 1) on X-chromosome thus visible on profile as probe binds to it;

with 1 normal allele in each, DNA profile of each parent will show a single band, corresponding to fragment that contains normal allele;

(b)(ii)

boy (C) inherited an X-chromosome carrying defective allele from mother (A) and Y chromosome from father (B);

due to large deletion of DNA sequence, complementary probe is unable to hybridise, thus not visible on profile;

(c)(i)

daughter (2) and mother;

as heterozygous (carriers), only 1 allele is mutated while the other is normal;

normal allele has recognition site for *HindIII* thus cleaves intron sequence giving one 3 kb band (visible on Southern blot) and 1 kb band;

however, when mutated in the other allele (defective allele), *HindIII* recognition site is missing, thus giving only 1 band i.e. 4 kb (visible on Southern blot);

(c)(ii)

(1) and (4) are males with only one X-chromosome and Y chromosome;

in which (1) has the normal allele on X-chromosome containing recognition site for *HindIII* giving two bands of which one is the 3 kb band seen on Southern blot;

male (4), allele is mutated, missing *HindIII* recognition site, giving rise to only one 4 kb band on Southern blot;

(2) and (3) are both females thus having two X-chromosomes;

(2) is a heterozygous carrier as one X-chromosome is mutated thus not having recognition site, giving rise to 4 kb band and the other allele being normal gives one 3 kb band;

(3) is a homozygous female as both alleles are normal thus both contain *HindIII* recognition sites thus giving 3kb bands;

(d)

due to single point mutation, there are other more nucleotides on target sequence that complementary probes that can still bind to adjacent to mutation site as compared to mutation of a short sequence;

sequence with/ without single base mutation will still show up as dark spots on X-ray film after autoradiography;

Question 2

(a)

a genetic disease from autosomal recessive mutation;

mutation by deletion of three base pairs in chromosome 7 (CFTR gene);

mutation occurred in gamete forming cells of parents;

an individual offspring inherits one defective chromosome 7 from each parent;

resulting in homozygous recessive CF alleles;

two carrier/heterozygous parents have a 25% chance of producing offspring with cystic fibrosis offspring;

(b)(i)

phospholipid bilayer of liposome mimic structure of cell surface membranes;

allows interaction and fusion with cell membranes allowing gene to enter via endocytosis;

easier to prepare than viral vectors;

can carry large size of DNA to be transferred;

(b)(ii)

restriction enzymes that act on restriction sites flanking the CFTR gene is used to digest human DNA;

DNA fragments are separated by agarose in gel electrophoresis;

separated DNA fragments are then transferred to a nitrocellulose filter by Southern Blotting;

radioactive gene probe is then used to hybridize with CFTR gene;

position of bound probe is determined by autoradiography;

DNA extracted from gel;

gene is amplified by polymerase chain reaction (PCR);

amplified gene is used for therapy by inserting into vector (e.g. viral, liposome etc);

(c)(i)

absence of 3 base pairs, CTT in individual Q;

(c)(ii)

missing 1 amino acid in polypeptide;

(c)(iii)

mutation of nucleotide bases via gene mutation;

deletion of 3 base pairs coding for an amino acid;

missing 1 amino acid in polypeptide;

affecting folding of CFTR protein due to missing R group in amino acid;

CFTR protein becomes dysfunctional;

(d)

frequency of mutant CF allele would increase;

effect of heterozygous advantageous trait;

mutant CF allele will be preserved in natural populations;

as higher chances of passing down to further generations;

Question 3

(a)(i)

maize with a foreign desirable gene integrated within its genome and is expressed;
foreign gene is from another organism, in this case a bacterium, *Bacillus thurigiensis*;

(a)(ii)

maize carrying *Bt* gene produces a toxin in its leaves which kills insects that attack them;
gives better yield of crop;
reduce use of insecticides which are harmful to humans / environment;

(b)

microinjection, gene gun, viral vectors;

(c)(i)

as number of days increases from Day 1 to 4, mean area of leaves eaten per caterpillar increases in (A);

from 0.05 to 0.58 arbitrary units for (A);

throughout all days, smaller leaf area is consumed by Group A compared to Group C (control);

from 0.2 to 1.6 arbitrary units in (C);

smaller increase in area of leaf consumed by Group A could be due to decreased survival as less caterpillars survive to eat the leaves;

from 100% to 55% survival of caterpillars in Group (A) while 100% throughout in (C);

(c)(ii)

caterpillars that consumed pollens from non-genetically modified were unaffected whereas % survival of caterpillars that consumed pollens from *Bt* maize decreased with time;

100% throughout 4 days for (B) and (C) while from 100% to 55% decrease in survival of caterpillars for (A);

(d)

population of Monarch butterflies would decrease;

feeding period of caterpillars coincide with shedding of maize pollens;

high percentage of caterpillars would consume pollens from Bt maize that landed on leaves of milkweed;

4 days after consuming *Bt* pollens, only 55% of the caterpillars survived;

high percentage of caterpillars that consumed *Bt* pollens would be killed;

Question 4

(a)

Recombination

1. isolate human gene (e.g. human insulin gene, anti-thrombin gene, human growth factor gene etc);
2. using a suitable restriction enzyme / endonuclease;
3. produce DNA fragment with complementary sticky ends;
4. select bacterial cells with plasmid DNA (vector) which contain antibiotic resistance genes;
5. use same restriction enzyme to digest plasmid DNA at antibiotic resistance gene marker;
6. plasmid cleaved forming complementary sticky ends;
7. mix DNA containing gene of interest, with plasmid DNA
8. reannealing of sticky ends via complementary base pairing of sticky ends;
9. DNA ligase is added to form catalyse formation of phosphodiester bonds to join restriction fragments;
10. human gene of interest inserted into vector forming recombinant plasmid/DNA;

Transformation

11. introduce recombinant plasmids into host cells e.g. *Escherichia coli*;
12. mix recombinant plasmids to a culture of *E. coli*;
13. subjected to calcium chloride followed by brief heat shock or electroporation;
14. holes appear in cell surface membranes making them permeable to DNA (competent cells);
15. recombinant plasmids enter bacterial host cells (transformed cells);

Selection

16. identification of transformed bacterial cells containing recombinant plasmids;
17. identify bacteria cells carrying recombinant plasmids by plating bacteria onto nutrient agar plate using velvet/nylon cloth;
18. via replica plating from master plate to replica plate containing antibiotic medium;
19. insertional inactivation – human gene of interest incorporated in antibiotic marker gene cause inactivation of marker gene;
20. select colonies that are not able to survive on replica plate from the master plate (bacteria containing recombinant plasmid);

Culturing

21. expression of human gene in bacteria cell to produce desired human protein;
22. cloning of bacteria cells containing recombinant plasmids;
23. a large population of bacteria, all containing inserted human gene, is produced after repeated divisions;
24. following replication, expression (transcription and translation) of human gene occurs within host cells to give rise to desired human protein;

Harvesting

25. multiple copies of bacterial cells formed via fermentation when subjected to optimal growth conditions;
26. human protein is harvested bursting the bacterial cells, isolated, extracted and purified for use;

(b)

1. **totipotent** embryonic stem cells, unspecialized;
2. from fertilized egg into 8-cell stage embryo develops into blastocyst via mitosis;
3. retains complete set of genetic information;
4. can give rise to any cell type / entire organism;
5. **pluripotent** embryonic stem cells from inner cell mass of blastocysts;
6. capable of continually renewing and dividing through cell division for long periods of time;
7. develops into three primary germ layers, ultimately giving rise to entire organism;
8. **multipotent** (adult) stem cells important in developing tissues that give rise to multiple specialised cell types;
9. multipotent haemopoietic stem cells of adult stem cells;
10. giving rise to all types of blood cells e.g. erythrocytes, B lymphocytes, T lymphocytes, neutrophils, basophils, platelets;
11. **unipotent** (adult) stem cells give rise to new cells of the same specific type, e.g. liver cells
12. small number of adult stem cells found in various parts of the body, e.g. in brain / bone marrow / blood vessels / skeletal muscle / skin / liver;
13. adult stem cells important as they generate replacements for cells that are lost in disease, tissue injury & normal wear-and-tear;
14. neuronal stem cells develops into neurons / nerve cells / skin stem cells develops into skin nerve cells / bone marrow stem cells develops into blood cells, liver cells, muscle cells;

(c)

Note:

Students are expected not only to state the points but also to elaborate briefly on them. They are expected to provide at least 2 ethical and 2 social implications.

1. genetically modified (GM) food might cause allergies;
2. GM food may prove to be toxic / carcinogenic;
3. genes from GM plants may transfer into the wild gene pool through cross-pollination;
4. may lead to reduction in biodiversity,
5. 'tampering with nature' / religious objections (ignore refs to 'playing God')
6. objections to consuming animal genes in plants by vegetarians and some religious groups;
7. developments may benefit rich countries at the expense of poorer countries as latter can afford expensive technology;
8. world food production may become dominated by small number of large companies;
9. additional stress/strain on animals to increase production of meat/milk etc; e.g. oncomouse, pharmsheep;
10. ownerships issues – unfair for large multinational companies to patent GM animals
11. violations to animal rights, cruelty to animals resulting in diseased and deformed animals, oncomouse – e.g. higher risk of disease;
12. biopiracy – foreign exploitation of natural resources;